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CHAPTER 2

Natural History, Ecology, and the Conservation and Management of Sea Otters

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This chapter introduces key aspects of the life history and ecology of sea otters. History—namely their near extinction from overhunting followed by protection and recovery—provides a context for much of what is known about sea otter ecology as well as an important point of reference for key issues having to do with their management and conservation. The species' unique form and function led to their value as an economically valuable resource during the fur trade, the keystone role they play in the ecology of coastal marine ecosystems, their often extreme human appeal, and the challenges they face as the world's smallest fully marine mammal. Although the sea otter is just one of the millions of species that deserve attention in the endeavor of biodiversity conservation, they highlight a number of issues that are central to achieving that goal.

Keywords

Bet-hedging; food web; keystone species; metabolism; Pacific maritime fur trade; shellfish; spatially structured population; trophic cascade

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Most if not all of our planet's 1.2 million described extant species (and the 10 million or more others that remain unknown or undescribed—Mora et al., 2011) have been affected by humankind's sundry and pervasive influences. These influences have resulted in a 100- to 1000-fold increase in extinction rates (Lawton and May, 1995) with the anticipated loss of roughly half of the extant species over the next 100 years (Wilson, 2002). Even our own species' survival to the end of the twenty-first century has been questioned (Rees, 2003).

Given the enormity of this situation and the fact that many species would seem to be in graver danger of impending extinction than are sea otters, why would an entire volume be devoted to the conservation of this one species? There are at least three reasons. First, the sea otters' high trophic status, requirement for relatively large areas of habitat, and keystone role (sensu Paine, 1969) in ecosystem function (Estes, 2008) make them umbrellas for the conservation of other associated species (Soulé and Terborgh, 1999). Second, sea otters conflict with several human interests and activities. In particular, they are vulnerable to environmental pollution (especially spilled oil; see Chapter 4), and they compete with people for shellfish resources. Last, sea otters command inordinate public attention simply because many people find them appealing (see Chapter 12).

Although this volume is explicitly about conservation (which I think of as the endeavor to preserve biodiversity and maintain the functional integrity of ecosystems), I am including management (which I think of as the manipulation of species or their habitats so as to enhance their benefits or reduce their costs to human societies) in the scope of my discussion because these endeavors are so intimately intertwined. I will begin by identifying the main issues surrounding the conservation and management of sea otters (Table 2.1). This list can be divided into those issues that have actually resulted in conservation or management actions

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and those for which actions have only been proposed or imagined. Both the understanding and resolution of these issues are defined to an important extent by the features of sea otters, the species with which sea otters interact, and the environments in which sea otters live. My purpose in this chapter is to identify these features and discuss why they are important to conservation and management.

Table 2.1

Issues and Actions Concerning the Conservation and Management of Sea Otters

For Which Actions Have Been Taken
• Protection from overharvest • Translocations to reestablish populations • Protection from incidental mortality in fisheries • Protection from terrestrial pathogens • Rehabilitation
Issues Not Yet Actionable
• Shellfish conflicts • Indirect effects of predation on other species and ecosystem processes • Pollution (chemical and biological) • Genetic diversity • Human disturbance • Resource economics and tourism

History

The present is inextricably linked to and thus an inevitable product of the past. A consideration of history is therefore integral to understanding the conservation and management of any species. In this section I will briefly recount the history of sea otters and their environment on two time scales—pre- and post-contact with nonindigenous peoples.

Where should a pre-contact history of sea otters begin? One might look back a half billion years to the Cambrian explosion, which marks the beginnings of metazoan evolution. That's far too long for my purposes and those of most people who will read this volume. Instead, I will begin in the late Miocene or early Pliocene, some 3 to 5 million years ago, at the onset of the current glacial age (Pielou, 1992). Low productivity tropical/subtropical conditions characterized higher latitude oceans of the northern hemisphere before that time. Increased production associated with the onset of polar cooling likely drew the sea otters' first marine ancestors from their primitive freshwater ecosystems into the coastal oceans of the North Pacific. A species not unlike the modern sea otter dispersed into the North Atlantic Ocean via the transarctic interchange (Willemsen, 1992), but subsequently became extinct. Otherwise, the *Enhydra* lineage (see Box 2.1 for a synopsis of nomenclature and phylogeny) has been restricted to the North Pacific Ocean and southern Bering Sea by frozen oceans to the north, New and Old World land masses to the east and west, and tropical conditions to the south.

Box 2.1

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Nomenclature

Common name. Sea Otter

Order: Carnivora

Family: Mustelidae

Scientific name: *Enhydra lutris*

Subspecies. *E.l. nereis* (California), *E.l. kenyoni* (Washington to Alaska), and *E.l. lutris* (Asia) (Wilson, 1991)

Phylogeny, Classification, and Population Structure

The modern sea otter is the only completely marine species in the family Mustelidae (skunks, weasels, minks, badgers, and honey badgers) (Wozencraft, 1993). The sub-specific taxonomy of the sea otter was based largely on morphological analyses (Wilson, 1991) that is at least partially supported by subsequent molecular genetic data. Scribner et al. (1997) and Larson (see Chapter 5) conclude that although populations today are differentiated genetically, limited sequence divergence and lack of phylogeographic concordance suggest an evolutionarily recent common ancestor and some degree of gene flow throughout their range.

As the earliest sea otters radiated into marine environments, they were confronted with a choice—whether to feed on shellfishes or finfishes. Seemingly they could not do both because of the different morphological and behavioral traits required for efficient exploitation of these two prey types. As it was, they became shellfish predators, possibly because competition with pinnipeds and other piscivores precluded them from exploiting finfishes or possibly simply because this is what their freshwater ancestors did. Whatever the reason, sea otters evolved to consume a wide array of benthic macroinvertebrates, including such groups as echinoderms, gastropods, bivalves, and crustaceans (Riedman and Estes, 1990; see also Box 2.2 for further details on diet). Sea otters developed effective means for the capture and consumption of these prey, such as forelimb strength and sensitivity, bunodont dentition for crushing the exoskeletons of their invertebrate prey, and tool use as an aid to breaking through the exoskeletal defenses of especially well-armored species such as bivalves and gastropods.

Box 2.2

Diet and Foraging Behavior

The sea otter is a generalist predator, known to consume more than 150 different prey species (Kenyon, 1969; Riedman and Estes, 1990; Estes and Bodkin, 2002). With few exceptions, their prey consist of sessile or slow moving benthic invertebrates such as mollusks, crustaceans, and echinoderms. Foraging occurs in habitats with rocky and mixed-sediment substrates between the high intertidal to depths slightly in excess of 100 m. Preferred foraging habitat is generally in depths less than 25 or 40 m (Riedman and Estes, 1990), although studies in southeast Alaska have found that some animals forage mostly at depths from 40 to 80 m (Bodkin et al., 2004; see Chapter 3).

The diet of sea otters is usually studied by observing prey items brought to the surface for consumption, and therefore diet composition is usually expressed as a percentage of all identified prey that belong to a particular prey species or type. Although the sea otter is known to prey on a large number of species, only

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a few tend to predominate in the diet in any particular area. Prey type and size depends on location, habitat type, season, and length of occupation (see Chapters 6 and 10 for further analyses of sea otter diets). In California, otters foraging over rocky substrates and in kelp forests mainly consume decapod crustaceans, gastropod and bivalve mollusks, and echinoderms (Ebert, 1968; Estes et al., 1981). In protected bays with mixed sediments, otters mainly consume infaunal clams (*Saxidomus nuttallii* and *Tresus nuttallii*; Kvitek and Oliver, 1988). Along exposed coasts of mixed sediments, the Pismo clam (*Tivela stultorum*) is a common prey (Stevenson, 1977). Important prey in Washington State include crabs (*Cancer* spp., *Pugettia* spp.), octopus (*Octopus* spp.), intertidal clams (*Protothaca* spp.), sea cucumbers (*Cucumaria miniata*), and the red sea urchin (*Strongylocentrotus franciscanus*) (Kvitek et al., 1989). The predominately mixed-sediment habitats of southeast Alaska, Prince William Sound, and Kodiak Island support populations of clams that are the primary prey of sea otters. Throughout most of southeast Alaska, burrowing clams (species of *Saxidomus*, *Protothaca*, *Macoma*, and *Mya*) predominate in the sea otter's diet (Kvitek et al., 1993). They account for more than 50% of the identified prey, although urchins (*S. droebachiensis*) and mussels (*Modiolis modiolis*, *Mytilus* spp., and *Musculus* spp.) can also be important. In Prince William Sound and Kodiak Island, clams account for 34–100% of the otter's prey (Calkins, 1978; Doroff and Bodkin, 1994; Doroff and DeGange, 1994). Mussels (*Mytilus trossulus*) apparently become more important as the length of occupation by sea otters increases, ranging from 0% at newly occupied sites at Kodiak to 22% in long-occupied areas (Doroff and DeGange, 1994). Crabs (*C. magister*) were once important sea otter prey in eastern Prince William Sound, but apparently have been depleted by otter predation and are no longer eaten in large numbers (Garshelis et al., 1986). Sea urchins are minor components of the sea otter's diet in Prince William Sound and the Kodiak archipelago. In contrast, diet in the Aleutian, Commander, and Kuril islands is dominated by sea urchins and a variety of finfish (including hexagrammids, gadids, cottids, perciformes, cyclopterids, and scorpaenids; Kenyon, 1969; Estes et al., 1982). Sea urchins tend to dominate the diet of low-density sea otter populations, whereas more fishes are consumed in populations near equilibrium density (Estes et al., 1982). For unknown reasons, fish are rarely consumed by sea otters in regions east of the Aleutian Islands.

Sea otters also exploit episodically abundant prey such as squid (*Loligo* spp.) and pelagic red crabs (*Pleuroncodes planipes*) in California and smooth lumpsuckers (*Aptocyclus ventricosus*) in the Aleutian Islands and the Gulf of Alaska. On occasion, sea otters attack and consume sea birds, including teal (*Anas crecca*), scoters (*Melanita perspicillata*), loons (*Gavia immer*), gulls (*Larus* spp.), grebes (*Aechmophorus occidentalis*), and cormorants (*Phalacrocorax* spp.) (Kenyon, 1969; Riedman and Estes, 1990).

Sea otters evolved into a species with strong limiting effects on their benthic invertebrate prey, in turn affecting many other species through indirect interactions. The kelps (Laminariales) for example, which also had a North Pacific center of origin (Estes and Steinberg, 1988), evolved in an environment in which their principal enemies—herbivorous sea urchins—were limited to cryptic habitats and held at small population sizes by prototypical sea otter predation (see Box 2.3 and food web section [below] for additional detail). This situation decoupled the potential for an evolutionary arms race (i.e., the co-evolution of defense and resistance) between kelps and their herbivores, in turn facilitating the development of a modern kelp flora that is poorly defended (chemically and morphologically) against herbivory (Steinberg et al., 1995). That co-evolutionary scenario might explain the extreme vulnerability of Pacific kelps to sea urchin grazing and the

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dramatic phase shifts between kelp forests and urchin barrens that have followed the loss or recovery of sea otters across much of the North Pacific Ocean (Estes and Duggins, 1995; Watson and Estes, 2011; Estes et al., 2013). It might also explain why the kelps, freed from the burden of investing resources into defenses against their herbivores, have become so highly productive (Reed and Brzezinski, 2009).

Box 2.3

Sea Otters, Sea Urchins, and Kelp—A Trophic Cascade

Sea otters eat sea urchins and sea urchins eat kelp. Joined together, these direct consumer–prey interactions create a positive indirect effect of sea otters on kelp (see illustrations and images below). Indirect interaction chains that begin with high trophic level species and radiate downward through the food web via top-down forcing processes are known as "trophic cascades" (Paine, 1980). This effect was discovered in kelp forest ecosystems by comparing islands with and without sea otters (created by the fortuitous survival of otters at some islands and their extinction at others following the maritime fur trade) in the western Aleutian archipelago (Estes and Palmisano, 1974). Subsequent research has shown this otter–urchin–kelp cascade to occur broadly and predictably across much of the North Pacific Ocean (Estes and Duggins, 1995; Kvitek et al., 1998; Watson and Estes, 2011). The biotic communities living on and around the shallow rocky reefs exist in two distinct phase states, one of which is characterized by an abundance of kelp and few urchins and the other of which is characterized by abundant urchins and few kelps. The transition between these phase states as otter densities change occurs as an abrupt phase shift (Steneck et al., 2002; Estes et al., 2010), which varies as a function of sea otter density with geographic region and depending upon whether otter populations are increasing or declining (i.e., the system displays "hysteresis," Scheffer, 2009). The mechanisms responsible for this behavior are now fairly well understood. When a system is in the kelp-dominated state, it tends to remain in that state because detrital fall-out from the overlying kelp canopy provides adequate nutritional resources for sea urchins without the urchins having to move in search of food. Under this circumstance the urchins have little or no destructive effect on the living kelps. Moreover, the kelp forest, once established, interacts with wave surge to physically repel invading sea urchins through a whiplash effect (Konar, 2000). When a system is in the urchin-dominated state, it also tends to remain there because the hungry urchins are continually on the move in search of food, thereby destructively grazing any newly settled kelp plant and preventing kelp from becoming established (Konar and Estes, 2003).

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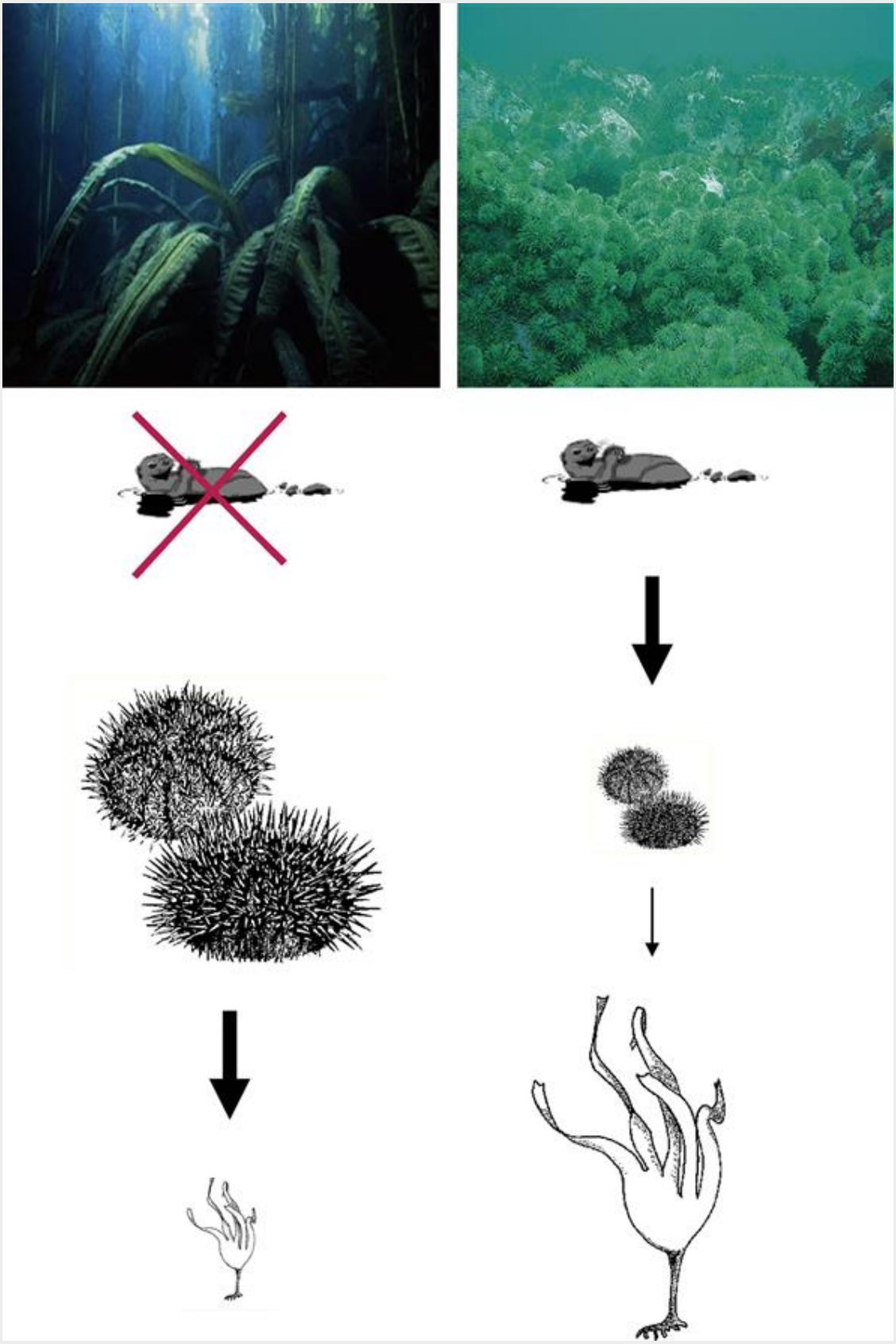

The drawing illustrates the mechanisms by which sea otter predation on sea urchins cascades downward through the kelp forest food web to create either kelp-dominated (left) or urchin-dominated (right) communities. Photos at the top are from islands in the western Aleutian archipelago with (left) and without (right) sea otters.

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Several other evolutionary influences of the "trophic cascade" (sensu Paine, 1980) from sea otters to urchins to kelps are known or suspected. For example, the late Cenozoic radiation of hydrodamaline sirenians (the lineage that ultimately led to Steller's sea cow [*Hydrodamalis gigas*]—a kelp consumer) from their tropical, sea grass-consuming ancestors into the North Pacific Ocean (Domning, 1978) must have been facilitated, if not primarily driven, by the great abundance and high nutritional quality of North Pacific kelps (Estes and Steinberg, 1988). Similarly, the diversity and large body size of North Pacific abalones may have been facilitated by the sea otter's evolutionary role in creating such a productive environment with high algal food quality (Estes et al., 2005). Although the direct effect of sea otters on abalone limits their size and abundance, the indirect effect through the otter–urchin–kelp trophic cascade promoted increased growth and body size (Figure 2.1). I suspect this evolutionary process played out not on the open sea floor but instead within the many substrate cracks and crevices that have provided safe haven to abalone from sea otter predation (Lowry and Pearse, 1973) since the Miocene.

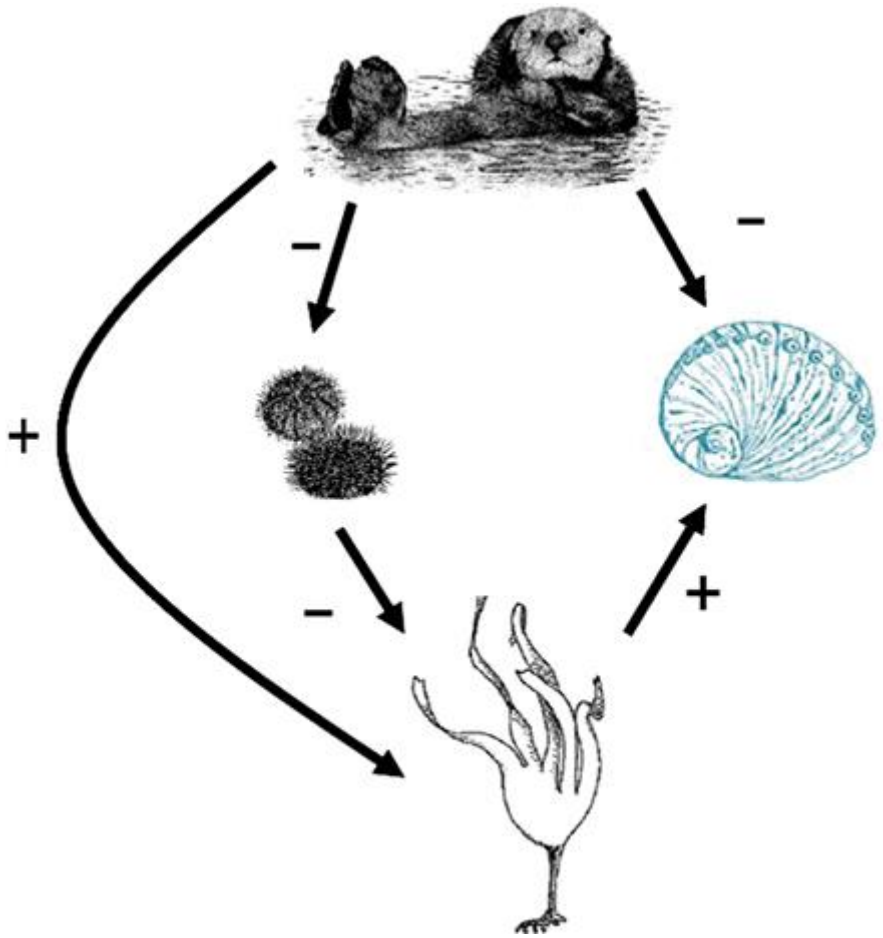

FIGURE 2.1 A food web diagram showing the potential conflicting influences of the direct (negative) and indirect (positive) influences of sea otters on abalones. See text for further explanations.

The post-contact history of sea otters began in 1741 with the Bering Expedition's discovery of an abundant otter population in the Commander Islands. This discovery initiated the Pacific maritime fur trade, which reduced once-abundant otter populations to near extinction over the ensuing 160 years (see Chapter 4). The quest for sea otter pelts strongly influenced patterns of human colonization and the political history of the Northeast Pacific region (Meinig, 1986). For instance, the depletion of sea otters was an integral factor underlying Russia's willingness to sell Alaska to the United States (Gibson, 1976). The near extinction of sea

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otters also caused the standing biomass of their benthic invertebrate prey to increase, which in turn fueled the development of coastal North Pacific commercial shellfisheries (Estes and VanBlaricom, 1985). Following protection of sea otters in the early 1900s and the regrowth of populations in more recent decades, conflicts with these fisheries have become a point of extreme contention and debate in the conservation and management of sea otters (see Chapters 11 and 12).

Form and Function

In this section I will briefly describe some of the sea otter's basic biology and discuss how several aspects of the species' form and function are integral to what I see as the three main dimensions of their conservation and management—exploitation and recovery, fishery conflicts, and marine pollution.

Insulation

Homeothermy at or around 37°C is a feature of all birds and mammals, demanding in turn a careful balance between heat production and heat loss. In contrast with pinnipeds and cetaceans, which use blubber for insulation against heat loss to the cold aquatic medium in which they live, sea otters depend entirely on their fur. The sea otter's exclusive dependency on fur promoted the evolution of their luxuriant pelt, thus leading to the species' demise in the Pacific maritime fur trade. Even today, the value of sea otter fur is the principal motivation for their exploitation by Alaska natives, again raising questions about the sustainability of this endeavor in some areas (Esslinger and Bodkin, 2009; Bodkin and Ballachey, 2010). The sea otter's reliance on fur for insulation also makes it especially vulnerable to the fouling effects of oil spills (Bodkin et al., 2002), which have increased in frequency in recent decades and will likely continue to pose a threat to sea otters so long as human societies extract fossil fuels and transport them across the oceans.

Metabolic Rate

High metabolic rate (Morrison et al., 1975; Davis et al., 1988; Costa and Kooyman, 1984; Yeates et al., 2007) is another feature of sea otter form and function that is germane to their conservation and management. This high metabolic rate is due in part to their small body size and the scaling of metabolic rate with body size (Kleiber, 1961; Brown et al., 2004), in part to their mustelid ancestry (which as a group has an elevated metabolism—Iverson, 1972; McNab, 1989), and in part to their need to generate heat for the maintenance of thermal neutrality in a cold aquatic environment. A high metabolic rate coupled with high population densities (compared with other mammalian carnivores of similar body size) underlies the conflict between sea otters and shellfisheries (see Chapters 4, 11, and 12).

Life History

Yet another important feature of sea otter form and function that is relevant to the species' conservation and management is life history strategy, or the manner in which they acquire and allocate resources to reproduction and survival in order to maximize fitness (Stearns, 1989). This strategy in sea otters is a product of two forces—the constraints of their evolutionary history and selection imposed by life in the sea. An understanding of the consequences of these forces can be gleaned by comparing sea otters with other

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mustelids (with whom they share a recent common ancestor), and with other marine mammals (with whom they share a common environment). Such a comparison reveals two important features of the sea otter's life history—litter size and reproductive tactics (Box 2.4).

Box 2.4

More on Sea Otter Life History

In addition to the limits placed by a single offspring, other traits limit population growth. Females usually attain sexual maturity at age 3–4 (Bodkin et al., 1993). von Biela et al. (2009) reported age of first reproduction varies by a year or less among populations, depending on the degree of resource limitation. Adult female reproductive rates are consistently high and range from 0.80 to 0.94 (Siniff and Ralls, 1991; Bodkin et al., 1993; Jameson and Johnson, 1993; Riedman et al., 1994; Monson and DeGange, 1995; Monson et al., 2000). Whereas reproductive output remains relatively constant over a broad range of ecological conditions, pup survival appears to be more strongly influenced by resource availability. At Amchitka Island, a population that was at or near equilibrium density, dependent pup survival ranged from 0.22 to 0.40, compared to nearly 0.85 at Kodiak Island, where food was not limiting and the population was increasing (Monson et al., 2000). Various factors can contribute to variation in pup survival, including female experience (Riedman et al., 1994; Monson and DeGange, 1995) and body condition (weight/unit length) (Bodkin and Monson, 2002; see Chapter 6). Post-weaning juvenile survival appears to be a primary mechanism of regulation in many sea otter populations, and post-weaning survival rates, which can vary annually from 0.18 to 0.86 (Monson et al., 2000; Ballachey et al., 2003), also appear to be affected by the status of the population relative to food availability. Factors affecting survival of young otters (both pre- and post-weaning) appear to be more important than reproductive rates in regulating abundance in many populations. Once a sea otter survives its first year of life, there appears to be a relatively high probability of surviving to senescence where density dependent mechanisms are structuring population abundance. Such may not be the case when density independent factors such as predation are important. Survival of sea otters more than 2 years of age is generally high, approaching 0.90, gradually declining over time (Bodkin and Jameson, 1991; Monson et al., 2000). Maximum ages, based on tooth annuli, are about 22 years for females and 15 years for males. A female at the Seattle Aquarium lived to be 27 years old.

Causes of mortality in sea otter populations are inherently difficult to determine, with the probability of detecting and assigning cause of death dependent on the cause. For example, the carcass of a sea otter that dies of starvation is more likely to be recovered than one killed by a predator. Documented sources of mortality include predation, starvation, disease, contaminants, incidental take by humans, intentional human harvest, and intraspecific aggression. Recognized sea otter predators include the white shark (*Carcharadon carcharias*), brown bear (*Ursus arctos*), wolf (*Canis lupus*), red fox (*Vulpes vulpes*), wolverine (*Gulo gulo*), killer whale (*Orcinus orca*), and bald eagle (*Haliaeetus leucocephalus*) (Kenyon, 1969; Ames and Morejohn, 1980; Riedman and Estes, 1990; Monson and DeGange, 1995; Hatfield et al., 1998; Bodkin, 2001).

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In contrast with all other mustelids (which have multiple young litters), sea otters are like marine mammals in having single-offspring pregnancies. This simple observation suggests that litter size in sea otters has been under strong selection by life in the sea. The net effect on their population biology is an r_{max} value of about 20% per year (Estes, 1990; see also Chapter 10 and Box 2.4). This one feature of the sea otter's life history limits the rate at which populations can be sustainably harvested and the rate at which depleted populations can recover (Bodkin and Ballachey, 2010).

Most marine mammals are extreme "investment strategists" (Stearns, 1989) in that they store energy by accumulating blubber during good times and expend this stored energy to fuel metabolism during bad times. These animals often feed in one part of their range and give birth to and nourish their young while living at an energy deficit elsewhere. Because individual females of such investment strategists can assess resource availability well in advance of giving birth (based on blubber storage), they have the innate capacity to make well-informed pre-partum (i.e., at the time of copulation and during pregnancy) decisions on whether or not to reproduce. The sea otters' high metabolic rate and inability to store energy defines the species as an extreme "income strategist," which means that they must feed regularly to survive and reproduce, and that they have no means of assessing pre-partum resource availability at the time of fertilization. Perhaps for this reason, sea otters have adopted a reproductive strategy known as "bet-hedging" (Cohen, 1966; Monson et al., 2000). That is, adult females become pregnant immediately after parturition and almost always carry their pregnancies to term. The reproductive decision is not whether to become pregnant but rather if or when to abandon their pups. Mothers apparently make that decision shortly following parturition. The combined effects of their income and bet-hedging strategies may thus account for the large number of small, abandoned pups, which in California are often discovered by the public and raised in captivity for later release to the wild.

Important dimensions to the conservation and management of sea otters have arisen from the combined effects of an extreme income strategy and their dependency on benthic invertebrates. These features hold the species close to shore where it has been easy to observe and study but where the animals also must endure a close association with human impacts. Their inability to store energy together with their nutritional dependency on benthic invertebrates has limited population recovery in some areas because of their inability to disperse across expanses of deep water (between islands, for example).

Associated Species

Sea otters associate and interact with numerous other species. In this section I will discuss how the characteristics of species in the coastal ecosystem help define the nature of species interactions, and how these interactions affect the conservation and management of sea otters. I will focus this discussion on three functional groups—autotrophs, free-living macroinvertebrates, and parasites—because of their importance to the way sea otters and their coastal ecosystems interact and operate.

Autotrophs

Coastal ecosystems contain two main groups of autotrophs—single-celled phytoplankton and multicellular macrophytes (algae and vascular plants). Together these organisms fuel production of coastal oceans across the sea otter's range.

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Phytoplankton have short generation times, which means that phytoplankton populations can grow and decline rapidly. Such changes are typically responses to physical forcing (e.g., wind, upwelling, changes in the strength and direction of currents) and grazing by herbivorous zooplankton. Phytoplankton-dominated systems often shift quickly between high and low rates of production. The influences of these shifts on sea otters and the influences of sea otters on phytoplankton through indirect species interactions are largely unknown.

Macroalgae (i.e., kelps [Laminariales], rockweeds [Fucales], and other groups) comprise what are often referred to as kelp forests. In contrast with phytoplankton, the interactions between sea otters and macroalgae have been extensively studied and are reasonably well known. Although the life span (and thus the generation time) of a typical macroalga (typically one to a few years) is much longer than that of a phytoplankter (typically days to weeks), it is short compared with those of terrestrial plants (commonly decades to centuries, especially in forest systems). Macroalgae also have complex life histories that involve gametophyte and sporophyte life stages. Both the spores and gametes are free-living and mobile, thus creating the possibility for rapid, long-range dispersal (Reed et al., 1988). Gametophytes function much as terrestrial seed banks (Edwards, 2000) during periods that are unfavorable for the larger and more conspicuous sporophyte life states. Macroalgal sporophytes often die rapidly and en masse over large areas, either because of unfavorable ocean conditions (i.e., large ocean waves or warm, nutrient-impoverished water—Dayton et al., 1992; Edwards and Estes, 2006) or intense grazing (Steneck et al., 2002). Most macroalgal species are susceptible to grazing and can appear or disappear from an area quickly in response to changes in the intensity of herbivory (Steneck et al., 2002).

Kelp forests also are characterized by high rates of primary production, owing to the fact that their essential resources—water, nutrients, and light—are seldom limiting. This high rate of primary production fuels high levels of secondary production through a detritus-based food web (Duggins et al., 1989).

Macroinvertebrates

Like macroalgae, most of the sea otter's invertebrate prey have complex life histories with large, sessile, or weakly motile adult stages and small, highly dispersive larval stages. These characteristics create the capacity for long-range dispersal, rapid but sometimes highly episodic recruitment, and high rates of population growth. Larval dispersal and recruitment are influenced by the vagaries of coastal ocean currents, and high rates of population growth in the post-settlement adults are fueled in significant part by macroalgae and the detritus-based food web. Many of the sea otter's macroinvertebrate prey are valued by human consumers, thus resulting in conflicts between sea otters and shellfisheries in some areas (see Chapters 4, 11, and 12).

Parasites and Pathogens

Like most species, sea otters are parasitized by a variety of viruses, bacteria, protozoa, and multicellular organisms. And like macroalgae and invertebrates, most of these forms have complex life histories with dispersive life stages. However, unlike nonparasitic species, many of these life stages are not free-living but instead occur within the bodies of their hosts. Mostly for this reason, the life stages of parasites and pathogens are cryptic and their population dynamics are difficult to study and understand. These difficulties are exacerbated for sea otters in the coastal oceans by potential linkages between the land and the sea. For example, the eggs of *Toxoplasma gondii*, a protozoal parasite that is nonreproductive but pathogenic in sea

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otters, are believed to infect sea otters through runoff from land to the adjacent coastal ocean (Conrad et al., 2005).

Food Web

A food web can be thought of as a roadmap of who is eaten by whom. Species are the elements of this map (analogous to towns and cities) and consumer–prey interactions are the linkages among the species (analogous to roads that connect the towns and cities). It is the nature of these linkages among species that determine the dynamic properties of food webs. Coastal food webs in general and the functional importance of sea otters to coastal food web structure and organization in particular have been extensively studied and are relatively well understood. My focus in this section will be on two properties of these coastal food webs: the linkages among species, and the resulting dynamical properties of populations and functional groups.

Consumer–prey interactions are by definition +/− in nature, which is to say that one species (the consumer) benefits while the other species (the prey) incurs a cost. The magnitude of this cost determines the *interaction strength* (Paine, 1992), defined as the difference in prey population size when the consumer is present or absent (Berlow et al., 1999). A second property of consumer–prey interactions is the extent to which they link up with one another to provide functional connectivity through the food web.

Sea otters are strong interactors in that they can have large negative effects on their prey populations. Sea urchins also are strong interactors because, when present in sufficient numbers, they have large negative effects on macroalgal populations. These two sets of strong interactions link together in kelp forest ecosystems to form a “trophic cascade” (Paine, 1980) in which sea otters exert an indirect positive effect on macroalgal populations (Estes and Palmisano, 1974; also see Box 2.3), in turn affecting other ecological processes and numerous associated species (Estes, 2005). Consumer–prey linkages also connect kelp forests with the open ocean (e.g., the episodic spawning migration of oceanic fishes to the coastal zone that in turn provide an important food resource to sea otters [Watt et al., 2000]; predation by the largely oceanic killer whales on sea otters [Estes et al., 1998]).

The aforementioned dynamical features of kelp forest food webs have important implications for sea otter conservation and management. Strong interactions between sea otters and their invertebrate prey have produced conflicts between sea otters and shellfisheries. However, follow-on linkages from otter–prey interactions to other species and ecosystem processes produce a broader array of influences by sea otters on coastal ecosystems, some of which are beneficial (e.g., kelp forest enhancement and related benefits to nearshore and oceanic fishes) to human welfare. Finally, the nonlinear dynamics of these consumer–prey interactions result in a system for which the critical transitions between phase states are abrupt and variable across regions in their functional forms (i.e., the way in which they vary with sea otter density).

Ecosystems

Ecosystems are communities of living organisms together with the nonliving (physical and chemical) elements of the surrounding environment. High productivity, a defining feature of higher latitude marine ecosystems, is accounted for by an abundance of the physical and chemical requisites for photosynthesis—water, nutrients, carbon, and light. These conditions result in the remarkably high rates of primary production by kelp forest ecosystems in general (Reed and Brzezinski, 2009) and by sea otter-dominated kelp

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forest ecosystems in particular (Duggins et al., 1989; Wilmers et al., 2012). Another possible contributing factor to high coastal production is the physical energy provided by tidally generated water movement. Although such kinetic energy is not in itself usable in biosynthesis, it does move large amounts of organic material across the sea floor where it can fuel secondary production in suspension-feeding organisms. This process may explain the high production and high associated sea otter densities in places like lower Cook Inlet and Glacier Bay in the northeast Gulf of Alaska, where tidal amplitudes and rates of tidally generated water flow are extreme.

A second important feature of coastal marine ecosystems is the diverse ways in which they are linked to land on one side, the open sea on the other, and the atmosphere above. Several of these linkages involve the consumer–prey interactions that were discussed earlier in the section on food web dynamics. However, land–sea–atmosphere linkages are more complex than this. The bulleted examples in Box 2.5 illustrate that complexity.

Box 2.5

Inter-Ecosystem Linkages

- Runoff vectors' sediments, nutrients, toxins, pathogens, and perhaps other materials from land into the coastal ocean.
- Coastal marine production can in turn facilitate terrestrial productivity, especially in places where nutrients or water are otherwise limiting (Polis and Hurd, 1996; Maron et al., 2006).
- The larvae of coastal invertebrates and fishes may spend weeks or months in the water column, sometimes at great distances from shore, before returning to the coastal ocean as recruits to adult populations (Roughgarden et al., 1988). Ocean currents may transport these dispersive life stages long distances from their parent stocks (Palumbi, 2003).
- The open sea can also subsidize coastal ocean predators and thus increase their impacts on coastal ecosystems (Estes et al., 1998; McCaughley et al., 2012).
- Atmospheric CO_2 diffuses into coastal ocean water where it acidifies the aquatic environment and is consumed by photosynthesis. The high rate of production by kelps serves to both sequester atmospheric carbon and to reduce ocean acidification (Reed and Brzezinski, 2009; Wilmers et al., 2012).

A third important characteristic of modern coastal ecosystems is their proximity to densely populated human settlements. The large numbers of people living near ocean margins use and modify these coastal environments in myriad ways.

The aforementioned features of coastal ecosystems influence their conservation and management in several obvious ways. For instance, their high productivity leads to the human exploitation of fish and other biological resources. The complex linkages among land, the coastal oceans, the open sea, and the atmosphere create an "openness" to the dynamics of all four systems that complicates the manner in which they work and thus the ease with which they can be conserved and managed. Some of the materials that are vectored by runoff from land to the coastal ocean constitute health risks to wildlife and humans.

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The highly productive coastal ocean together with the sea otter's inability to feed beyond the shallow coastal zone is responsible for remarkably high sea otter population densities in some areas. The species' polygynous mating system, which is characterized in part by male territoriality, probably evolved because of these high population densities and the resulting ability of males to compete for females. In this regard sea otters are more like pinnipeds (Bartholomew, 1970) than other species of otters (Estes, 1989).

Because sea otters are carnivores, high population densities depend not only on high net primary productivity (NPP) but also on an effective trophic linkage between that NPP and animals that are suitable prey. Since most of the sea otter's prey species are invertebrates with larval life histories and open populations, their abundance and sustainability as a sea otter prey resource depends on larval delivery and settlement. Regional variation in larval delivery (probably caused by local oceanographic processes) might thus cause sea otter population densities to vary in ways that are not directly linked to NPP.

Human Emotions

Decisions concerning the conservation and management of natural resources are ultimately driven by two factors—logic and emotion. Sea otters engender strong emotions by both their advocates and detractors. The emotional responses to sea otters by their detractors (nearly all of whom are people who value the shellfish that otters eat) are fairly straightforward and easy to understand. The same cannot be said for those of their advocates.

Sea otters are unarguably among the world's most appealing and charismatic animals. Many people love them in ways they could never love a worm or an insect or even a fish. This emotional reaction, which is entirely visceral in nature, has led to what I see as a dichotomy of values that revolves around the love of animals on the one hand and the love of nature on the other. A love of animals places high value on the welfare of individuals, just as we relate to our family, friends, and pets. A love of nature places value instead on the preservation of species, biodiversity, and the functional integrity of ecosystems. These values aren't necessarily mutually exclusive but they often lead to conflicting views of and objectives for conservation and management. Such a scenario may be unfolding in California where there is an ongoing effort to save, rehabilitate, and release orphaned sea otter pups back to the wild population (see Chapter 9). This effort is seen by many as complementary to the goal of recovering the threatened southern sea otter population and maintaining or restoring the functional integrity of its ecosystem. However, others see the rehabilitation of orphaned sea otter pups and the conservation of sea otter populations as being at cross purposes. The argument by this latter group is that the orphaned pups that were destined to die (because of the sea otter's basic reproductive strategy and their mothers' choice to abandon them—as explained previously in the section on function and form), intraspecific competition is an important limiting influence on the California sea otter population, and thus rehabilitating and returning orphaned pups to nature may be detrimental to the wild animals that were otherwise destined to live. The other is that such efforts to rehabilitate orphaned sea otter pups divert time, attention, and resources from the conservation and management of sea otter populations and their associated communities and ecosystems.

Discussion

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It would be convenient if the approach to wildlife conservation and management was similar to that of most other science-based endeavors, such as building a bridge or designing a piece of software. But it isn't. Both latter activities can turn to well-established principles and working algorithms, which designers and engineers use with relative ease and a high degree of confidence to achieve their desired goals. General methods and models for conservation and management are in their infancy and most of those that do exist cannot be employed with a high degree of confidence that they will really work. The difference is that the infrastructural complexity of ecosystems is vastly greater than that of human constructs. It is for this reason that wildlife conservationists and managers have been forced to turn to the details of the ecology and natural history for guidance on the species or systems they are trying to conserve and manage.

The purpose of this chapter was to explore some of the linkages between ecology and natural history, conservation and management of sea otters, and coastal marine ecosystems. Toward that end, we have seen how a view of history was essential to understanding why the sea otter became what it is today and how the species functions as a keystone predator. The historical perspective helps us understand why fisheries conflicts have arisen as an inevitable consequence of the near extinction and recovery of sea otters. We have seen how the sea otter's life in a cold aquatic environment, its small body size, and its dependence on fur have together led to an extreme income strategy, which in turn affects the life history tactics employed by sea otters to maximize their fitness and lifetime reproductive success. And we have seen how this life history strategy has led to bet-hedging and pup abandonment as a means of population regulation and how the rehabilitation of these abandoned pups has taken sea otter conservation and management in a direction that appeals to human emotions but is not necessarily in the best interest of populations and their ecosystems.

The understanding of sea otter natural history and ecology adds dimension and depth to our thinking about two especially important aspects of sea otter management—the consequences of harvest and the criteria for delisting depleted populations. For example, the same extreme income strategy that prevents long-range dispersal and holds individual sea otters to small home ranges creates extreme spatial structure in populations, in turn making the species more vulnerable to exploitation than it would otherwise be if individuals moved and intermingled over large areas. The other side to this aspect of the sea otter's spatial ecology, which is of potential human benefit, is the increased ability to manage populations through selective harvest or removal over small spatial scales, if ever that is deemed necessary and desirable. Similarly, our understanding of the sea otter's keystone role in kelp forest ecosystem structure raises interesting questions about the targets and goals of conservation and management (such as the establishment of delisting criteria under the US Endangered Species Act). The maintenance of viable populations is presently the gold standard for conservation targets (Soulé, 1987; Morris and Doak, 2002). While that approach makes sense at one level, species do not live in isolation from their surrounding communities and ecosystems and we have seen that a minimum viable population is not sufficient to maintain the sea otter's keystone role in kelp forest ecosystems (Estes et al., 2010). The same argument holds in the establishment of conservation targets for any strongly interacting species (Soulé et al., 2003, 2004).

Much of what I have written reflects my own experiences and beliefs. Someone else, given the same charge, might have written something very different. Part of the reason for this is that much has been left unsaid. As reflected by the chapter's title, I have endeavored to establish the more important connections between

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ecology and natural history on the one hand and the conservation and management of sea otters and their coastal ecosystems on the other. The degree to which I have succeeded in this undertaking will be evident from the extent to which the various points and arguments surface again in the chapters that follow.

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